



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Summer 2025-2026

Section: L

Group No: 04

Project Report On AIUB BUS MANAGEMENT SYSTEM

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Submitted By:

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Obtained Marks for CO2 and CO4 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)		Total =	Evaluation Criteria (CO4)		Total =
Requirement fulfillment			Design & Innovation		
Validation			Teamwork & Ethics		
Verification			Communication		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are

	implementation of verification processes, and data flow is not considered.	incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	attention to ensuring data integrity and appropriate data flow within the system.	comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO4: Perform and solve the proper system functional requirements which combines several aspects of the work to display and verify a real-life based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Design & Innovation	Fails to identify project requirements or propose any feasible solution. No meaningful design, model, prototype, or simulation is presented.	Demonstrates limited understanding of project requirements. Proposed solutions are incomplete or lack feasibility, and the design or prototype is poorly developed.	Identifies basic project requirements and develops a simple solution. A functional model or simulation is presented but lacks completeness, innovation, or refinement.	Clearly formulates project requirements, evaluates alternative solutions, and develops a functional prototype or simulation that effectively addresses the problem with minor limitations.	Demonstrates exceptional ability to analyze requirements, propose and justify innovative alternative solutions, and develop a well-designed, reliable, and fully functional prototype or simulation that effectively solves the problem.
Teamwork & Ethics	Fails to contribute effectively to the team or disregards professional and ethical responsibilities.	Demonstrates limited collaboration and inconsistent individual contribution. Professional or ethical practices are only partially followed.	Participates in team activities and fulfills assigned responsibilities with acceptable professionalism and ethical conduct.	Collaborates effectively, makes significant individual contributions, and consistently demonstrates professional and ethical behavior throughout the project.	Demonstrates exemplary teamwork, leadership, accountability, and consistent adherence to professional, ethical, and academic integrity standards while making outstanding individual contributions.
Communication	Fails to communicate project outcomes effectively. The report, presentation, or	Demonstrates limited communication skills. The report lacks organization, the	Communicates project outcomes adequately through a structured report, acceptable	Presents a well-organized technical report, delivers a clear and professional presentation, and	Demonstrates exceptional technical communication through a comprehensive

	viva is incomplete or unsatisfactory.	presentation is unclear, and responses during the viva show insufficient understanding.	presentation, and satisfactory responses during the viva.	confidently defends the project during the viva with minor weaknesses.	report, engaging and professional presentation, and confident, well-supported responses during the viva, reflecting thorough understanding of the project.
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Table of Contents:

Page no.

1. Chapter: 01 (Introduction, Case Study) -----	6 -7
2. Chapter: 02 (User Story) -----	8-9
3. Chapter: 03a (ER Diagram) -----	10
4. Chapter: 03b (Use Case Diagram) -----	11
5. Chapter: 03c (Class Diagram) -----	12
6. Chapter: 03d (Data dictionary) -----	13-14
7. Chapter: 03e (Normalization) -----	15-17
8. Chapter: 03f (SQL Queries, Database Schema) -----	18-21
9. Chapter: 04 (Screenshots of forms with figure names) -----	22-25

AIUB Bus Management System

Introduction:

The **AIUB Bus Management System** is a Windows-based application developed to improve the management and coordination of university transportation services. Managing students, drivers, buses, routes, pickup locations, schedules, and transportation fees manually can be time-consuming and may lead to errors or inconsistencies. The proposed system provides a centralized platform where transportation-related information can be stored, managed, and accessed efficiently according to the responsibilities of different users.

The system is designed around three main user roles: **Admin, Student, and Driver**. The Admin is responsible for managing students, drivers, administrators, routes, pickup locations, bus assignments, and time slots. The Admin can assign a specific bus and driver to a particular route and time slot, monitor transportation activities, and generate reports related to bus assignments and fee collection. This helps the administration maintain accurate and organized transportation records.

The Student module allows students to manage their transportation-related activities after logging into the system. Students can select their preferred pickup location, route, and available time slot based on the bus assignments provided by the Admin. They can also view their transportation fee information, make payments, and update their profile details. The pickup selection process is connected to the available bus assignments, ensuring that students can only select transportation options that are available for their chosen time slot.

The Driver module provides drivers with access to information related to their assigned transportation schedules. Drivers can view their assigned buses, routes, dates, and time slots. They can also access the pickup locations and the list of students associated with their assigned schedule. In addition, drivers can confirm student pickups, allowing the system to maintain records of daily transportation activities and student boarding information.

The system includes **user authentication and role-based access control** to ensure that each user can access only the features relevant to their assigned role. Admin, Student, and Driver users have separate responsibilities and access permissions within the system. All necessary information is stored in a structured database containing user accounts, student profiles, driver profiles, routes, route locations, time slots, bus assignments, student pickup selections, payment information, and travel records.

The main objective of the **AIUB Bus Management System** is to reduce manual work, minimize errors, improve coordination, and maintain transportation-related information in a centralized and organized manner. By connecting bus assignments, routes, time slots, pickup locations, student selections, payments, and travel records, the system provides a more efficient approach to managing university transportation services.

Overall, the system provides a convenient and structured platform for the university administration, students, and drivers. It simplifies daily transportation-related tasks while ensuring that important information is properly

maintained and easily accessible to authorized users. Through this system, the management of university bus services can become more organized, reliable, and efficient.

Case-Study:

The **AIUB Bus Management System** was developed based on the daily transportation needs of students, drivers, and the university administration. At a university like AIUB, a large number of students depend on buses to travel between their homes and the university. Managing bus routes, pickup locations, drivers, schedules, and student information manually can become difficult, especially when many students use the service every day. The system was therefore designed to bring these activities together in one organized platform.

The system has three main types of users: **Admin, Student, and Driver**. Each user has different responsibilities. The Admin manages the overall transportation system, including student and driver information, administrator accounts, routes, pickup locations, buses, time slots, and bus assignments. The Admin can assign a particular bus and driver to a specific route and time slot for a selected date. This makes it easier to organize the daily bus schedule and keep track of which driver and bus are responsible for each trip.

Students use the system to choose their transportation preferences. After logging in, a student can select an available time slot, route, and pickup location. The available routes depend on the bus assignments made by the Admin, so students can select transportation options that are actually scheduled for that time. Pickup locations can also have different fares. Students can view their outstanding transportation fees and make payments through the system. They can also update their basic profile information when necessary.

Drivers use the system to see the transportation assignment given to them. They can view their assigned bus, route, date, and time slot, along with the pickup locations and the students who have selected those locations. During the journey, the driver can confirm when a student has been picked up. This information is recorded in the system and helps maintain a record of the daily transportation activities.

The system also handles user accounts and access according to different roles. Student and driver accounts can be managed by the administration, while the system ensures that users can access the features appropriate to their roles. The system also keeps records of bus assignments, student pickup selections, payments, and travel activities. These records help the administration review the transportation service and generate useful reports.

The database contains several related entities, including Users, StudentProfiles, DriverProfiles, AdminProfiles, Routes, RouteLocations, TimeSlots, Buses, BusAssignments, StudentPickupSelections, and TravelLogs. Primary keys are used to identify individual records, while foreign keys connect related information between the tables. For example, a route can have multiple pickup locations, a bus assignment connects a route with a driver and time slot, and a student's pickup selection connects the student with a pickup location and time slot. Travel logs are used to maintain records of daily transportation activities, including student pickup confirmations.

Overall, the system represents a more practical way of handling the university's bus transportation activities. Instead of keeping separate records and depending on manual communication, the administration, students, and

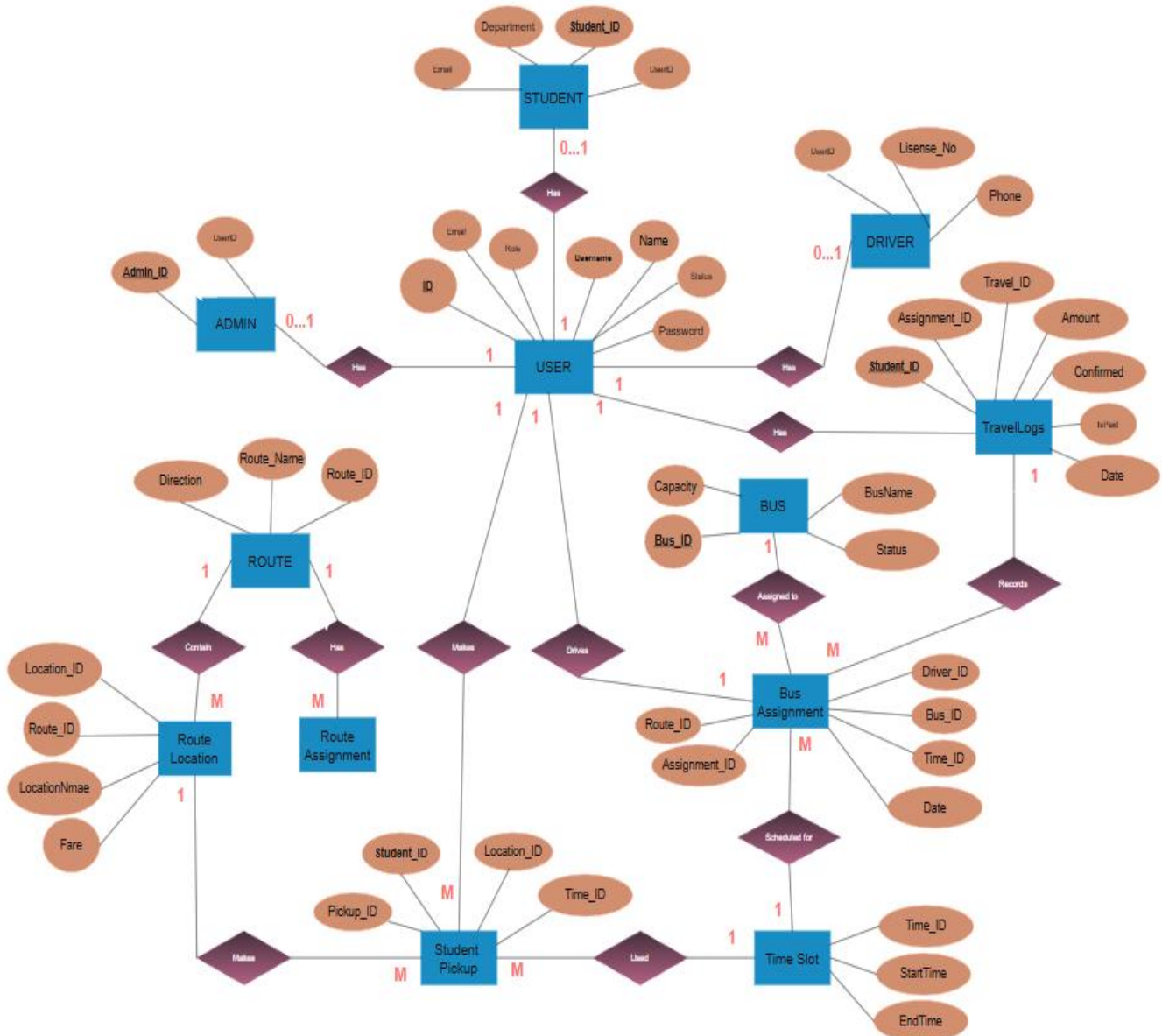
drivers can work with the same organized system. This makes daily bus management easier, keeps transportation information properly recorded, and provides students and drivers with the information they need for their scheduled trips.

Users Stories:

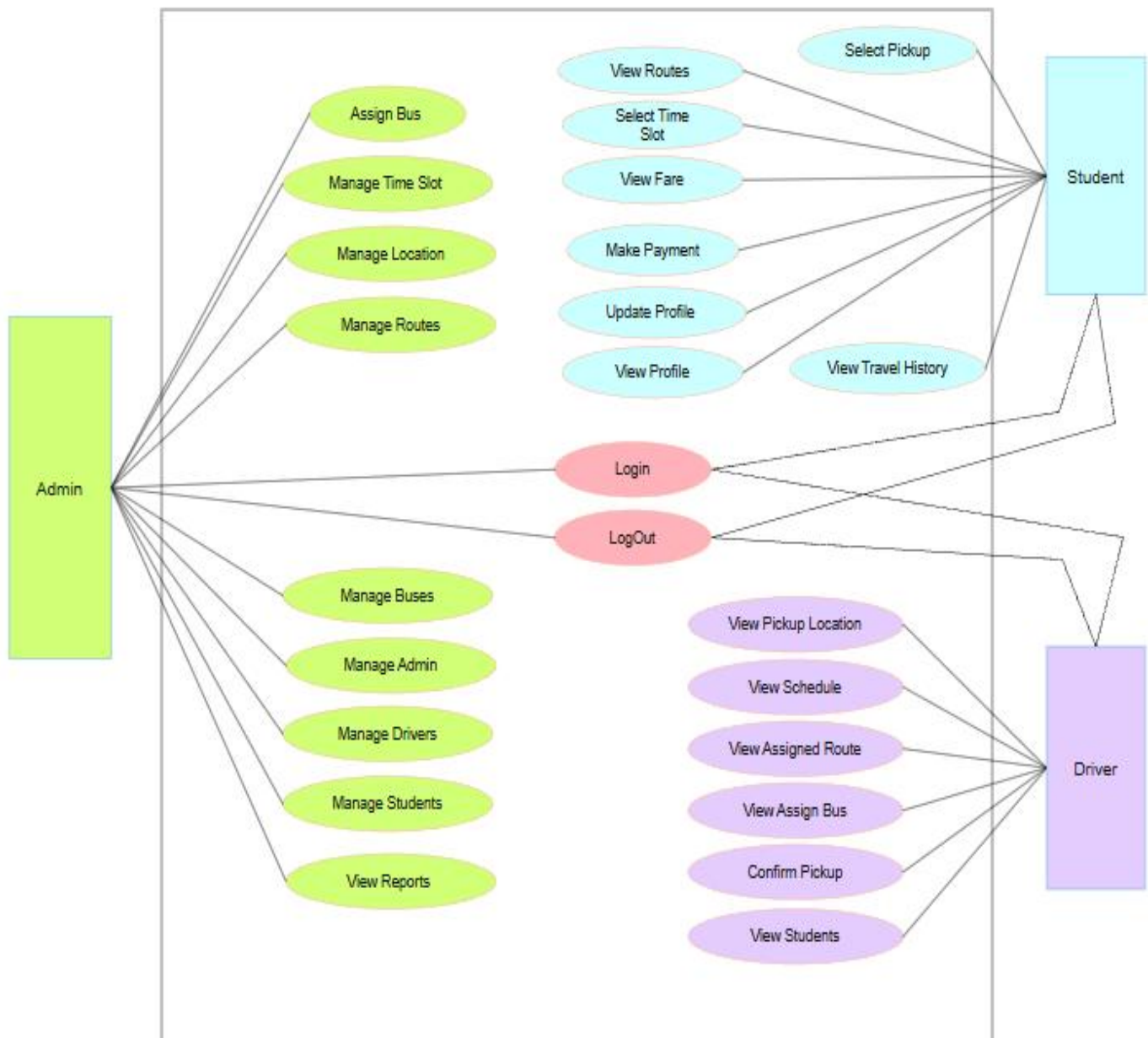
Use Case Name	Actor	User Story
Login	Student	As a student, I want to log in to the system so that I can access my transportation information and services.
View Profile	Student	As a student, I want to view and update my profile information so that my personal information
Select Time Slot	Student	As a student, I want to view available time slots so that I can choose a suitable transportation schedule.
Select Route	Student	As a student, I want to view available routes for my selected time slot so that I can choose an available bus route.
Select Pickup Location	Student	As a student, I want to select a pickup location so that I can choose where I will board the bus.
View Fare	Student	As a student, I want to view the assigned bus information so that I know which bus is available for my trip.
View Due Fees	Student	As a student, I want to view my outstanding transportation fees so that I know how much I need to pay.
Pay Transportation Fee	Student	As a student, I want to pay my transportation fee so that my payment can be recorded in the system.
Change Password	Student	As a student, I want to change my password so that I can keep my account secure.
Forgot Password	Student	As a student, I want to reset my forgotten password so that I can regain access to my account.
Login	Admin	As an admin, I want to log in to the system so that I can manage the university bus transportation system.

Manage Students	Admin	As an admin, I want to view, search, activate, and block student accounts so that student records remain organized and controlled..
Manage Routes	Admin	As an admin, I want to create and manage routes so that available transportation routes are properly maintained.
Manage Pickup Locations	Admin	As an admin, I want to add and manage pickup locations and fares so that students can choose suitable pickup points.
Manage Time Slots	Admin	As an admin, I want to create and manage time slots so that bus schedules can be organized properly.
View Reports	Admin	As an admin, I want to view transportation and payment reports so that I can monitor the overall bus service and collected fees.
Login	Driver	As a driver, I want to log in to the system so that I can access my assigned transportation information.
View Assigned Bus	Driver	As a driver, I want to view my assigned bus so that I know which bus I need to operate.
View Assigned Route	Driver	As a driver, I want to view my assigned route and time slot so that I know the route and schedule I need to follow.
View Pickup Locations	Driver	As a driver, I want to view the pickup locations for my assigned route so that I know where students need to be picked up.
View Students	Driver	As a driver, I want to view the students assigned to each pickup location so that I know which students I need to pick up.
Confirm Student Pickup	Driver	As a driver, I want to confirm when a student has been picked up so that the pickup activity is recorded in the system.
View Travel Records	Driver	As a driver, I want to view my transportation activity so that I can keep track of my assigned trips.
Change Password	Driver	As a driver, I want to change my password so that I can keep my account secure.

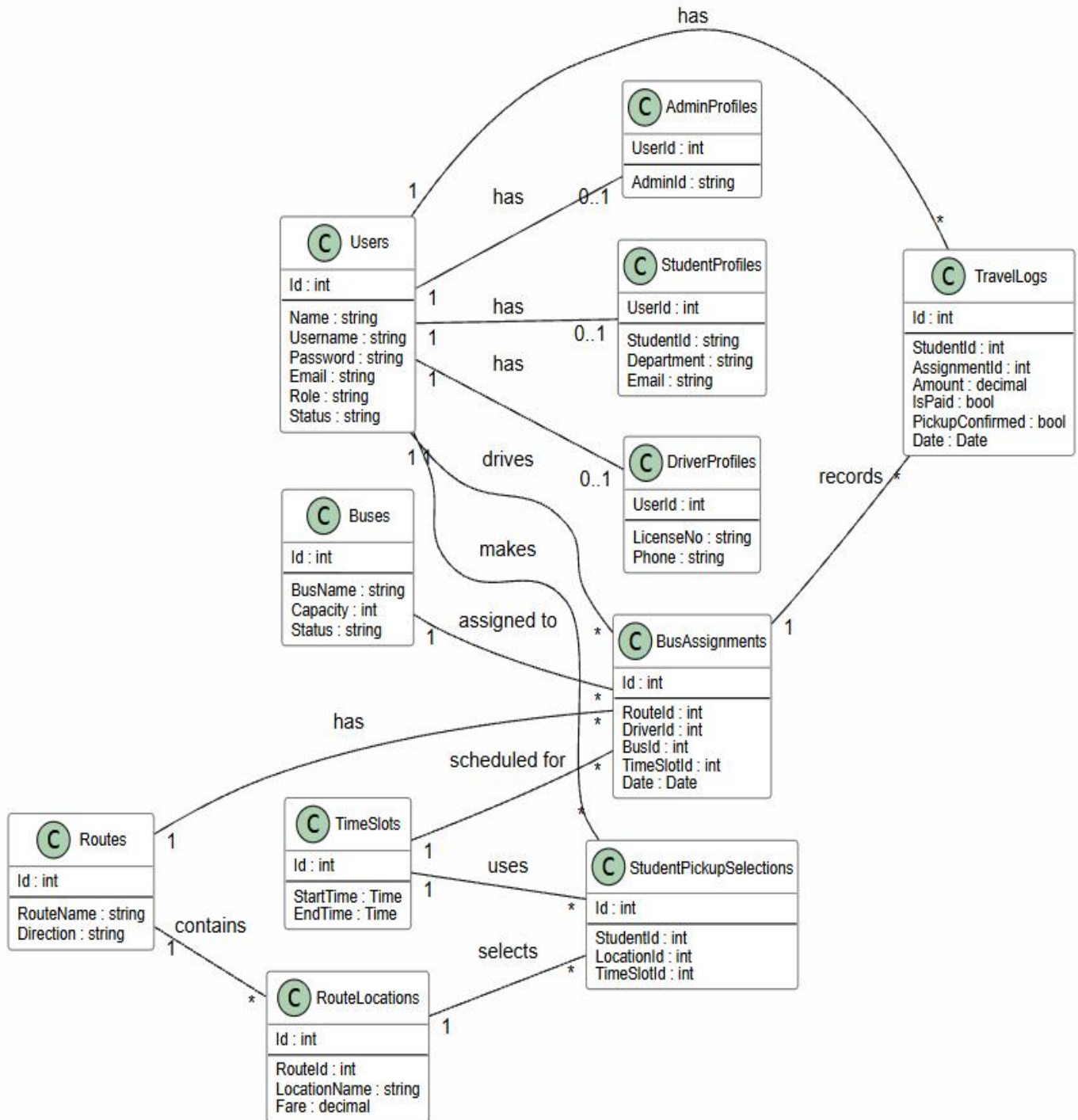
Er Diagram



Use Case Diagram



Class Diagram



Data Dictionary

AdminProfiles

Key	Name	Data Type	Nullable
PK, FK	UserId	Int	No
-	AdminId	varchar(50)	No

BusAssignments

Key	Name	Data Type	Nullable
PK	Id	Int	No
FK	RouteId	Int	No
FK	DriverId	Int	No
FK	TimeSlotId	Int	No
-	Date	Date	No
FK	BusId	Int	No

Buses

Key	Name	Data Type	Nullable
PK	Id	Int	No
-	RouteId	varchar(50)	No
-	Capacity	Int	No
-	Status	varchar(20)	No

DriverProfiles

Key	Name	Data Type	Nullable
PK, Fk	UserId	Int	No
-	LicenseNo	varchar(50)	No
-	Phone	varchar(20)	No

RouteLocations

Key	Name	Data Type	Nullable
PK	Id	Int	No
FK	RouteId	Int	No
-	Phone	varchar(100)	No
	Fare	decimal(10, 2)	NO

Route

Key	Name	Data Type	Nullable
PK	Id	Int	No
-	RouteName	varchar(50)	No
-	Direction	varchar(30)	No

StudentPickupSelections

Key	Name	Data Type	Nullable
PK	Id	Int	No
FK	StudentId	Int	No
FK	LocationId	Int	No
FK	TimeSlotId	Int	NO

StudentProfiles

Key	Name	Data Type	Nullable
PK, Fk	Id	Int	No
-	StudentId	varchar(30)	No
-	Department	varchar(100)	No
	Email	varchar(100)	NO

Timeslots

Key	Name	Data Type	Nullable
PK	Id	Int	No
-	StartTime	time(7)	No
-	EndTime	time(7)	No

TravelLogs

Key	Name	Data Type	Nullable
PK	Id	Int	No
FK	StudentId	Int	No
FK	AssignmentId	Int	No
	Amount	decimal(10, 2)	No
	IsPaid	bit	No
	PickupConfirmed	bit	No
	Date	Date	No

Users

Key	Name	Data Type	Nullable
PK	Id	Int	No
-	Name	varchar(100)	No
-	Username	varchar(50)	No
	Password	varchar(100)	No
	Email	varchar(100)	No
	Role	varchar(20)	No
	Status	varchar(20)	No

Normalization

1. User – Admin Profile

UNF:

User_ID, User_Name, Username, Password, Email, Role, Status, Admin_ID

1NF:

User_ID, User_Name, Username, Password, Email, Role, Status, Admin_ID

2NF:

(1) Users

User_ID(PK), User_Name, Username, Password, Email, Role, Status

(2) AdminProfiles

User_ID(PK, FK), Admin_ID

2. User – Student Profile

UNF:

User_ID, User_Name, Username, Password, Email, Role, Status, Student_ID, Department, Student_Email

1NF:

User_ID, User_Name, Username, Password, Email, Role, Status, Student_ID, Department, Student_Email

2NF:

(1) Users

User_ID(PK), User_Name, Username, Password, Email, Role, Status

(2) StudentProfiles

User_ID(PK, FK), Student_ID, Department, Email

3. User – Driver Profile

UNF:

User_ID, User_Name, Username, Password, Email, Role, Status, License_No, Phone

1NF:

User_ID, User_Name, Username, Password, Email, Role, Status, License_No, Phone

2NF:

(1) Users

User_ID(PK), User_Name, Username, Password, Email, Role, Status

(2) DriverProfiles

User_ID(PK, FK), License_No, Phone

4. Route – Route Location

UNF:

Route_ID, Route_Name, Direction, Location_ID, Location_Name, Fare

1NF:

Route_ID, Route_Name, Direction, Location_ID, Location_Name, Fare

2NF:

(1) Route

Route_ID(PK), Route_Name, Direction

(2) RouteLocations

Location_ID(PK), Route_ID(FK), Location_Name, Fare

5. Manages Bus

UNF:

Bus_ID, Bus_Name, Capacity, Status

1NF:

Bus_ID, Bus_Name, Capacity, Status

2NF:

(1) Bus

Bus_ID(PK), Bus_Name, Capacity, Status

6. Uses Time Slot**UNF:**

TimeSlot_ID, Start_Time, End_Time

1NF:

TimeSlot_ID, Start_Time, End_Time

2NF:

(1) TimeSlots

TimeSlot_ID(PK), Start_Time, End_Time

7. Bus – Route Assignment**UNF:**

Assignment_ID, Bus_ID, Bus_Name, Route_ID, Route_Name, Direction, Date

1NF:

Assignment_ID, Bus_ID, Bus_Name, Route_ID, Route_Name, Direction, Date

2NF:

(1) BusAssignments

Assignment_ID(PK), Bus_ID(FK), Route_ID(FK), Driver_ID(FK), TimeSlot_ID(FK), Date

(2) Bus

Bus_ID(PK), Bus_Name, Capacity, Status

(3) Route

Route_ID(PK), Route_Name, Direction

8. Driver – Bus Assignment**UNF:**

Driver_ID, Driver_Name, License_No, Phone, Assignment_ID, Bus_ID, Route_ID, TimeSlot_ID, Date

1NF:

Driver_ID, Driver_Name, License_No, Phone, Assignment_ID, Bus_ID, Route_ID, TimeSlot_ID, Date

2NF:

(1) DriverProfiles

Driver_ID(PK, FK), License_No, Phone

(2) BusAssignments

Assignment_ID(PK), Driver_ID(FK), Bus_ID(FK), Route_ID(FK), TimeSlot_ID(FK), Date

9. Student – Pickup Selection**UNF:**

Student_ID, Student_Name, Department, Location_ID, Location_Name, TimeSlot_ID, Start_Time, End_Time

1NF:

Student_ID, Student_Name, Department, Location_ID, Location_Name, TimeSlot_ID, Start_Time, End_Time

2NF:

(1) StudentProfiles

Student_ID, Department, Email, User_ID(FK)

(2) StudentPickupSelections

Selection_ID(PK), Student_ID(FK), Location_ID(FK), TimeSlot_ID(FK)

(3) RouteLocations

Location_ID(PK), Route_ID(FK), Location_Name, Fare

(4) TimeSlots

TimeSlot_ID(PK), Start_Time, End_Time

10. Student – Travel Log

UNF:

Student_ID, Student_Name, Assignment_ID, Route_ID, Bus_ID, TimeSlot_ID, Amount, IsPaid, PickupConfirmed, Date

1NF:

Student_ID, Student_Name, Assignment_ID, Route_ID, Bus_ID, TimeSlot_ID, Amount, IsPaid, PickupConfirmed, Date

2NF:

(1) TravelLogs

TravelLog_ID(PK), Student_ID(FK), Assignment_ID(FK), Amount, IsPaid, PickupConfirmed, Date

(2) BusAssignments

Assignment_ID(PK), Bus_ID(FK), Route_ID(FK), Driver_ID(FK), TimeSlot_ID(FK), Date

11. Student – Route / Pickup

UNF:

Student_ID, Student_Name, Department, Location_ID, Location_Name, Route_ID, Route_Name, Direction, TimeSlot_ID

1NF:

Student_ID, Student_Name, Department, Location_ID, Location_Name, Route_ID, Route_Name, Direction, TimeSlot_ID

2NF:

(1) StudentPickupSelections

Selection_ID(PK), Student_ID(FK), Location_ID(FK), TimeSlot_ID(FK)

(2) StudentProfiles

User_ID(PK, FK), Student_ID, Department, Email

(3) RouteLocations

Location_ID(PK), Route_ID(FK), Location_Name, Fare

(4) Routes

Route_ID(PK), Route_Name, Direction

(5) TimeSlots

TimeSlot_ID(PK), Start_Time, End_Time

Database Query Operations

1. Advanced SQL Query

```
SELECT
    r.RouteName,
    rl.LocationName,
    COUNT(sps.Id) AS StudentCount
FROM Routes r
INNER JOIN RouteLocations rl
ON r.Id = rl.RouteId
LEFT JOIN StudentPickupSelections sps
ON sps.LocationId = rl.Id
GROUP BY
    r.RouteName,
    rl.LocationName
ORDER BY
    r.RouteName,
    StudentCount DESC;
```

Explanation: This query uses JOIN, COUNT(), GROUP BY and ORDER BY to determine the number of students selecting each pickup location for every route.

Output Screenshot:

	RouteName	LocationName	StudentCount
1	AIUB - Uttara	uttara Sector 7	0
2	Badda - AIUB	Rampura	0
3	Badda - AIUB	Badda	0
4	Badda - AIUB	Bashundhara	0
5	Badda - AIUB	Aftabnagar	0
6	Badda - AIUB	Merul Badda	0
7	Jatrabari - AIUB	Malibagh	0
8	Jatrabari - AIUB	Mouchak	0
9	Jatrabari - AIUB	Mugda	0
10	Jatrabari - AIUB	Jatrabari	0
11	Jatrabari - AIUB	Sayedabad	0
12	Mirpur - AIUB	Mirpur-10	1
13	Mirpur - AIUB	Mirpur-1	0

2. Joins

```
SELECT
    ba.Id AS AssignmentId,
    b.BusName,
    r.RouteName,
    r.Direction,
    u.Name AS DriverName,
    ts.StartTime,
    ts.EndTime,
    ba.Date
FROM BusAssignments ba
INNER JOIN Buses b
ON b.Id = ba.BusId
INNER JOIN Routes r
ON r.Id = ba.RouteId
INNER JOIN Users u
ON u.Id = ba.DriverId
INNER JOIN TimeSlots ts
ON ts.Id = ba.TimeSlotId
ORDER BY
    ba.Date,
    ts.StartTime;
```

Explanation: This query demonstrates multiple INNER JOIN operations to combine bus, route, driver and time-slot information for each bus assignment.

Output Screenshot:

	AssignmentId	BusName	RouteName	Direction	DriverName	StartTime	EndTime	Date	
1	6	BUS-01	Uttara - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-12	
2	7	BUS-02	Mirpur - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-12	
3	8	BUS-03	Badda - AIUB	ToUniversity	karim	09:00:00.0000000	11:00:00.0000000	2026-09-12	
4	10	BUS-01	Mirpur - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-13	
5	13	BUS-01	Uttara - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-13	
6	14	BUS-01	Uttara - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-13	
7	21	BUS-01	Mohammadpur - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-13	
8	24	BUS-04	Jatrabari - AIUB	ToUniversity	karim	11:00:00.0000000	13:00:00.0000000	2026-09-13	
9	16	BUS-05	Jatrabari - AIUB	ToUniversity	karim	15:00:00.0000000	17:00:00.0000000	2026-09-13	
10	18	BUS-03	Badda - AIUB	ToUniversity	karim	15:00:00.0000000	17:00:00.0000000	2026-09-13	
11	9	BUS-01	Mirpur - AIUB	ToUniversity	karim	07:00:00.0000000	09:00:00.0000000	2026-09-14	
12	15	BUS-05	Jatrabari - AIUB	ToUniversity	karim	15:00:00.0000000	17:00:00.0000000	2026-09-14	
13	11	BUS-05	Jatrabari - AIUB	ToUniversity	karim	15:00:00.0000000	17:00:00.0000000	2026-09-14	

Query executed successfully. (localdb)\MSSQLLocalDB (17... DIDAR-RASUL\Acer (53) AIUBBusManagen

3. Subquery

```
SELECT
    Name,
    Username,
    Email
FROM Users
WHERE Id IN (
    SELECT StudentId
    FROM TravelLogs
```

Explanation: This query uses a subquery to retrieve users who have travel records in the TravelLogs table.

Output Screenshot:

150 %			
Results Messages			
	Name	Username	Email
1	didar rasul	didar	NULL

4. Dynamic Control Creation at Run-Time

```
private void CreateDynamicButtons(){
    // Remove previously created controls
    panelLocations.Controls.Clear();

    // Sample locations
    string[] locations =
    {
        "Uttara",
        "Abdullahpur",
        "Airport",
        "Banani",
        "Mirpur"
```

```

};

int top = 20;

foreach (string location in locations)
{
    Button btnLocation = new Button();

    btnLocation.Text = location;
    btnLocation.Width = 180;
    btnLocation.Height = 40;
    btnLocation.Left = 20;
    btnLocation.Top = top;

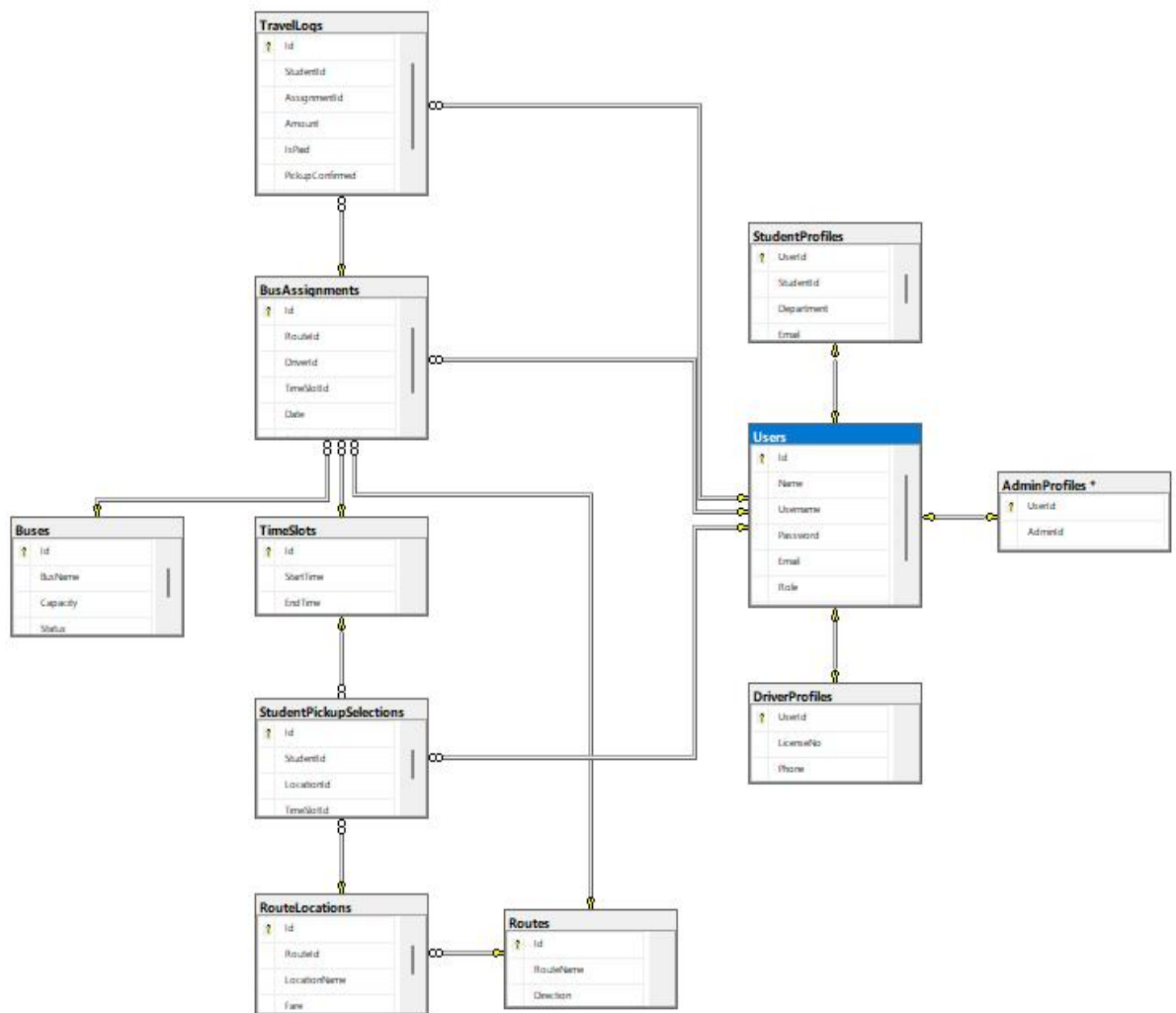
    panelLocations.Controls.Add(btnLocation);

    top += 50;
}
}

```

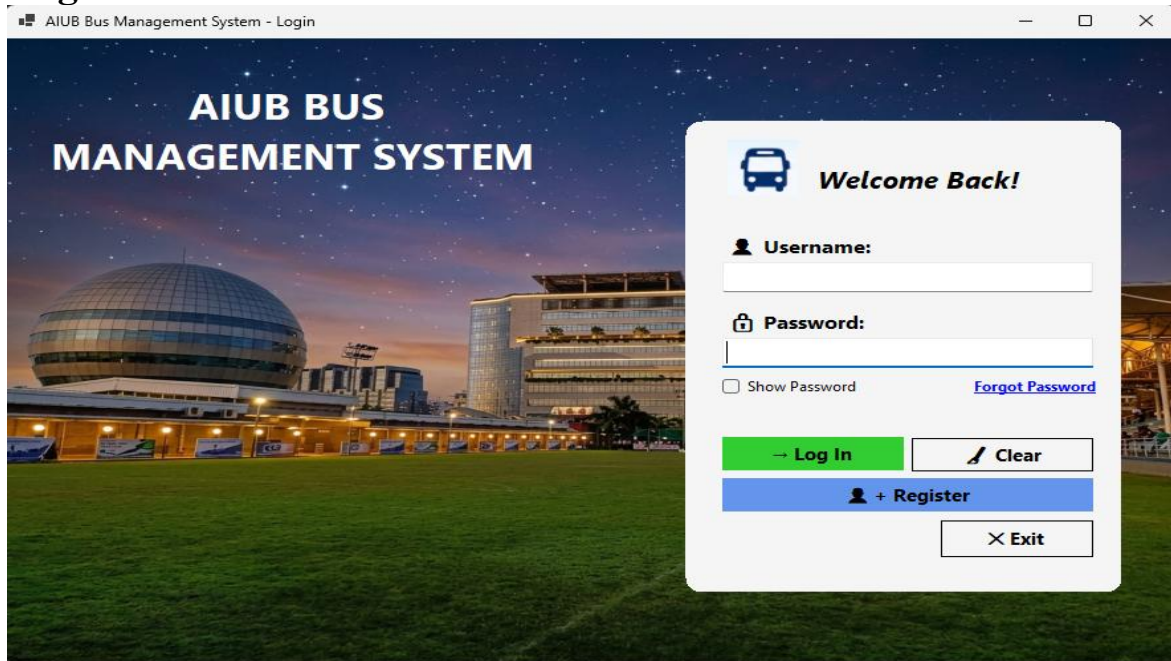
Explanation: This code dynamically creates RadioButton controls at run-time according to the pickup locations retrieved from the database. The controls are added to the group box without being permanently created in the form designer.

Dtabase Schema



Screenshots of forms with figure names

LogIn forms:



The login form is titled "AIUB BUS MANAGEMENT SYSTEM" and "Welcome Back!". It features a background image of the AIUB building at night. The form includes fields for Username and Password, a "Show Password" checkbox, a "Forgot Password" link, and buttons for "Log In", "Clear", "+ Register", and "Exit".

AIUB BUS MANAGEMENT SYSTEM

Welcome Back!

Username:

Password:

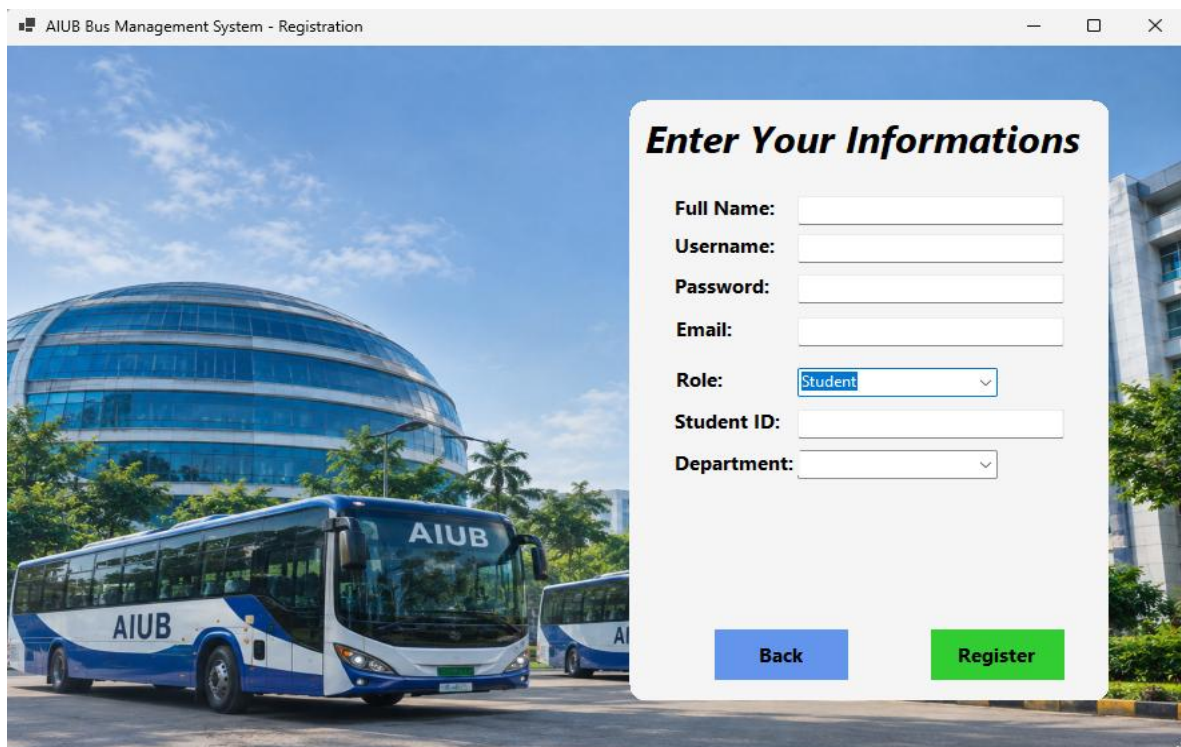
☐ Show Password [Forgot Password](#)

Log In Clear

+ Register

Exit

Registration forms:



The registration form is titled "Enter Your Informations". It features a background image of the AIUB building and a bus. The form includes fields for Full Name, Username, Password, Email, Role (with a dropdown menu), Student ID, and Department (with a dropdown menu). It also has "Back" and "Register" buttons.

Enter Your Informations

Full Name:

Username:

Password:

Email:

Role: Student

Student ID:

Department:

Back Register

Registration forms:

Forgot Password



Smart • Safe • Connected

Forgot Password

Username:

New Password:

Confirm Password:

Reset **Cancel**

Admin Forms (Main Dashboard) :

AIUB Bus Management System

Admin**Change password****Logout**

Manage Routes**Manage Students****Manage Driver****Assign Bus****Manage Admin****Reports**

Route Name **Direction** **Create Route**

	RouteName	Direction
▶	Uttara - AIUB	ToUniversity
	Mirpur - AIUB	ToUniversity
	Mohammadpur - AIUB	ToUniversity
	Badda - AIUB	ToUniversity
	Jatrabari - AIUB	ToUniversity

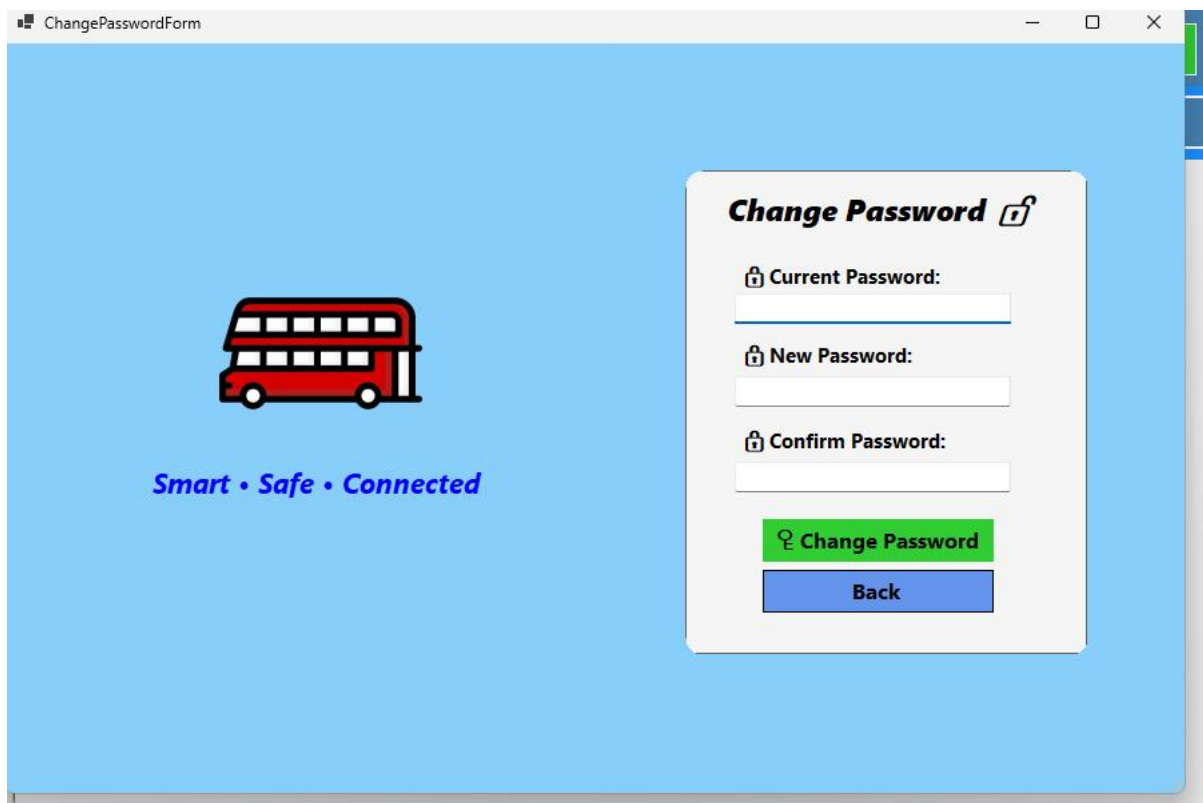
	LocationName	Fare
▶	Uttara Sector 7	50.00
	Uttara Sector 10	45.00
	House Building	40.00
	Airport	30.00
	Kuril	20.00

Location Name

Fare

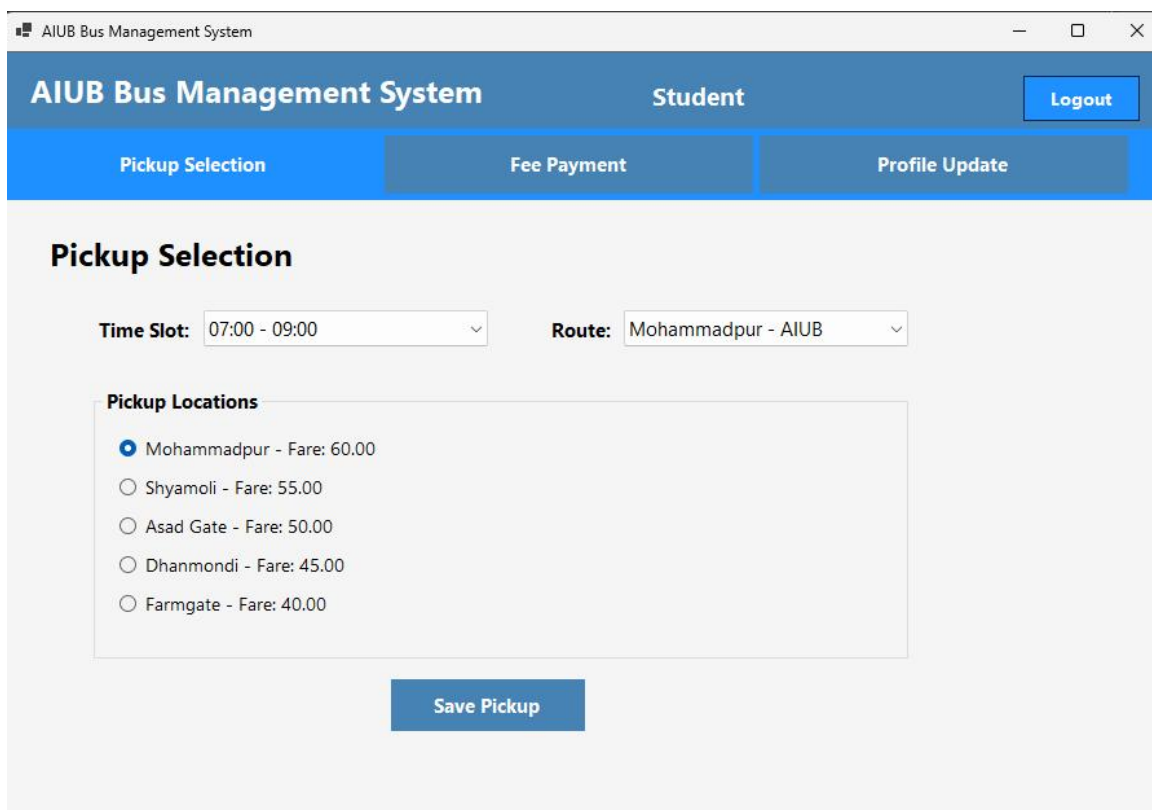
Add Location

Change Password Forms:



The screenshot shows a web browser window titled "ChangePasswordForm" with a light blue background. On the left, there is a red double-decker bus icon and the text "Smart • Safe • Connected" in blue. On the right, there is a white form titled "Change Password" with a lock icon. The form contains three password input fields labeled "Current Password:", "New Password:", and "Confirm Password:". Below these fields are two buttons: a green "Change Password" button and a blue "Back" button.

Student Forms (Main Dashboard) :



The screenshot shows the "AIUB Bus Management System" Student Dashboard. The header includes the system name, the user role "Student", and a "Logout" button. Below the header is a navigation bar with three buttons: "Pickup Selection" (highlighted in blue), "Fee Payment", and "Profile Update". The main content area is titled "Pickup Selection" and contains two dropdown menus: "Time Slot:" set to "07:00 - 09:00" and "Route:" set to "Mohammadpur - AIUB". Below these is a section titled "Pickup Locations" with a list of five locations, each with a radio button and a fare: "Mohammadpur - Fare: 60.00" (selected), "Shyamoli - Fare: 55.00", "Asad Gate - Fare: 50.00", "Dhanmondi - Fare: 45.00", and "Farmgate - Fare: 40.00". At the bottom of the form is a blue "Save Pickup" button.

Driver Forms (Main Dashboard) :

AIUB Bus Management System

AIUB BUS MANAGEMENT SYSTEM

Driver

Change Password

Log Out

Driver: karim

Assignments

	Id	Date	BusName	RouteName	Direction	TimeSlot	
▶	20	9/15/2026	BUS-01	Mohammadpu...	ToUniversity	07:00 - 09:00	
	22	9/15/2026	BUS-04	Jatrabari - AIUB	ToUniversity	11:00 - 13:00	
	23	9/15/2026	BUS-04	Jatrabari - AIUB	ToUniversity	11:00 - 13:00	
	25	9/15/2026	BUS-04	Jatrabari - AIUB	ToUniversity	11:00 - 13:00	

Pickup Locations

	LocationId	LocationName	StudentCount
	14	Asad Gate	0
	15	Dhanmondi	0
▶	16	Farmgate	0
	12	Mohammadpur	0
	13	Shyamoli	0
*			

Students

	StudentName	StudentId	PickupStatus	StudentUserId	TravelLogId
*					

Mark Picked